LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: The Fundamental Unit of Life (Cell)

Chapter / Topic: The Cell (topic-based)

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SECTION A

Q1. The nucleus of the cell was discovered by: **[1]**

Answer: (d) Robert Brown (1831)

Q2. Which organelle is called the 'kitchen of the cell' in plants? **[1]**

Answer: (c) Chloroplast (photosynthesis takes place here)

Q3. Which of these is absent in animal cells? **[1]**

Answer: (b) Cell wall

Q4. The movement of water molecules through a selectively permeable membrane is called: **[1]**

Answer: (a) osmosis

Q5. Which organelle is involved in forming lysosomes? **[1]**

Answer: (b) Golgi apparatus

Q6. **Assertion (A):** Mitochondria are able to make some of their own proteins. **Reason (R):** Mitochondria have their own DNA and ribosomes. **[1]**

Answer: (a)

SECTION B

Q7. Why is the plasma membrane called a selectively permeable membrane? **[2]**

Answer: It allows some materials to enter and leave the cell but stops others. In this way it controls what moves in and out of the cell.

- Allows some, stops others: 1
- Controls movement in/out: 1

OR, answer: Endocytosis is the taking in of food and other material from the outside by the flexible cell membrane. Example: Amoeba engulfs its food this way.

- Definition: 1
- Example: 1

Q8. Name two organelles that have their own genetic material (DNA). **[2]**

Answer: Mitochondria and plastids.

- Each: 1

SECTION C

Q9. Name the cell part that: **[3]**

(i) Smooth endoplasmic reticulum (SER). [1]
(ii) Endoplasmic reticulum (RER makes proteins; SER makes lipids). [1]
(iii) Leucoplasts (colourless plastids). [1]

Q10. Give three differences between plant cells and animal cells. **[3]**

Answer: Plant cells have a cell wall outside the cell membrane; animal cells do not. Plant cells have plastids (e.g. chloroplasts); animal cells do not. Plant cells usually have one large central vacuole, while animal cells have small vacuoles or none. (Centrosomes are found in animal cells.)

- Any three: 1 each

OR, answer: (a) Water enters the cell by osmosis and it swells. (b) Water leaves the cell and it shrinks. (c) There is no net movement of water, so the cell stays the same size.

- Each: 1

SECTION D

Q11. Answer the following: **[5]**

(i) The nucleus has a double-layered nuclear membrane with pores, and it contains chromosomes. Chromosomes are made of DNA and protein and become visible as rod-shaped structures when the cell divides. DNA carries the information for inheritance (genes). The nucleus controls the cell's activities and plays a central role in cell division. [3]

- Structure (membrane, chromosomes): 1
- DNA carries genetic information: 1
- Controls activities / cell division: 1

(ii) Mitosis: one cell divides into two identical daughter cells with the same number of chromosomes. It is used for growth and repair. Meiosis: occurs in the reproductive organs and gives four cells with half the number of chromosomes. It is used to form gametes. [2]

- Mitosis: 1
- Meiosis: 1

SECTION E

Q12. A student removed the shell of an egg by dissolving it in dilute hydrochloric acid, leaving the egg covered only by its thin membrane. The egg was… **[4]**

(i) Water moved into the egg through the membrane by osmosis, because the water outside was more dilute (hypotonic). [1]

(ii) Water moved out of the egg into the more concentrated (hypertonic) salt solution. [1]

(iii) Osmosis, through a selectively (semi-) permeable membrane. Examples (any one): raisins swelling in water; wilted vegetables becoming fresh in water; pickles shrinking in salt. [2]

- Osmosis + selectively permeable: 1
- Everyday example: 1